WORKED SOLUTION

STEP III 2013, Question 3

Pure • Vectors and the Sphere

STEP III 2013, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q3

The key idea

Everything flows from two facts: each $\mathbf{p}_i$ is a unit vector, and the four of them add to $\mathbf{0}$. Expanding a squared length as a dot product turns $(XP_i)^2$ into something *linear* in $\mathbf{x}\cdot\mathbf{p}_i$, and sums of linear terms collapse because $\sum \mathbf{p}_i = \mathbf{0}$. For the fourth powers a quadratic term survives, so we need actual coordinates, which is exactly what part (ii) provides.

Solution

The dot products $\mathbf{p}_i \cdot \mathbf{p}_j$

Since the tetrahedron is regular, all six edges have the same length. For $i \neq j$,

$$(P_iP_j)^2 = (\mathbf{p}_i - \mathbf{p}_j)\cdot(\mathbf{p}_i - \mathbf{p}_j) = |\mathbf{p}_i|^2 + |\mathbf{p}_j|^2 - 2\,\mathbf{p}_i\cdot\mathbf{p}_j = 2 - 2\,\mathbf{p}_i\cdot\mathbf{p}_j,$$

using $|\mathbf{p}_i| = 1$ (each vertex lies on the unit sphere). Equal edge lengths therefore force every $\mathbf{p}_i \cdot \mathbf{p}_j$ $(i \neq j)$ to take the same value; call it c.

Now take the dot product of $\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 + \mathbf{p}_4 = \mathbf{0}$ with $\mathbf{p}_1$:

$$\mathbf{p}_1\cdot\mathbf{p}_1 + \mathbf{p}_1\cdot\mathbf{p}_2 + \mathbf{p}_1\cdot\mathbf{p}_3 + \mathbf{p}_1\cdot\mathbf{p}_4 = 0 \quad\Longrightarrow\quad 1 + 3c = 0,$$

so $c = -\frac{1}{3}$. That is, $\mathbf{p}_i\cdot\mathbf{p}_j = -\frac{1}{3}$ for all $i \neq j$, as required.

Part (i)

Since $|\mathbf{x}| = 1$ too,

$$(XP_i)^2 = (\mathbf{p}_i - \mathbf{x})\cdot(\mathbf{p}_i - \mathbf{x}) = |\mathbf{p}_i|^2 - 2\,\mathbf{x}\cdot\mathbf{p}_i + |\mathbf{x}|^2 = 2 - 2\,\mathbf{x}\cdot\mathbf{p}_i.$$

Summing over i and using the distributive law for the dot product,

$$\sum_{i=1}^{4}(XP_i)^2 = 8 - 2\,\mathbf{x}\cdot(\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 + \mathbf{p}_4) = 8 - 2\,\mathbf{x}\cdot\mathbf{0} = 8,$$

as required.

Part (ii)

With $\mathbf{p}_1 = (0,0,1)$ and $\mathbf{p}_2 = (a,0,b)$, the condition $\mathbf{p}_1 \cdot \mathbf{p}_2 = -\frac{1}{3}$ gives $b = -\frac{1}{3}$ at once. Then $|\mathbf{p}_2| = 1$ gives

$$a^2 = 1 - b^2 = \tfrac{8}{9}, \qquad \text{so} \qquad a = \frac{2\sqrt{2}}{3},$$

taking the positive root since $a > 0$.

Let $\mathbf{p}_3 = (u,v,w)$. Three conditions pin it down:

$$\mathbf{p}_1 \cdot \mathbf{p}_3 = -\tfrac{1}{3} \implies w = -\tfrac{1}{3},$$

$$\mathbf{p}_2 \cdot \mathbf{p}_3 = -\tfrac{1}{3} \implies \tfrac{2\sqrt{2}}{3}u + \tfrac{1}{9} = -\tfrac{1}{3} \implies u = -\tfrac{4}{9} \cdot \tfrac{3}{2\sqrt{2}} = -\frac{\sqrt{2}}{3},$$

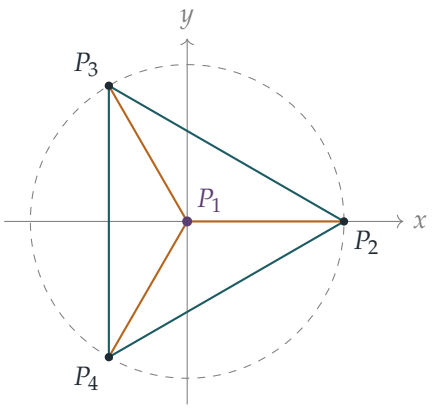
$$|\mathbf{p}_3| = 1 \implies v^2 = 1 - \tfrac{2}{9} - \tfrac{1}{9} = \tfrac{6}{9} \implies v = \pm\frac{\sqrt{6}}{3}.$$

Exactly the same equations hold for $\mathbf{p}_4$, so P_3 and P_4 are the two points this gives (they must be different points, so one takes each sign). Labelling them one way round,

$$P_3 = \Big(-\frac{\sqrt{2}}{3}, \frac{\sqrt{6}}{3}, -\frac{1}{3}\Big), \qquad P_4 = \Big(-\frac{\sqrt{2}}{3}, -\frac{\sqrt{6}}{3}, -\frac{1}{3}\Big).$$

(The labels of P_3 and P_4 can be swapped.) As a check, the four position vectors do add to $\mathbf{0}$, and $\mathbf{p}_3 \cdot \mathbf{p}_4 = \frac{2}{9} - \frac{6}{9} + \frac{1}{9} = -\frac{1}{3}$.

Seen from above, P_2, P_3, P_4 form an equilateral triangle in the plane $z = -\frac{1}{3}$, with P_1 directly above its centre:



Part (iii)

From part (i), $(XP_i)^2 = 2(1 - \mathbf{x} \cdot \mathbf{p}_i)$, so squaring,

$$(XP_i)^4 = 4(1 - \mathbf{x} \cdot \mathbf{p}_i)^2 \qquad \implies \qquad \sum_{i=1}^{4}(XP_i)^4 = 4\sum_{i=1}^{4}(1 - \mathbf{x} \cdot \mathbf{p}_i)^2,$$

as required. Expanding the bracket and using $\sum \mathbf{x}\cdot\mathbf{p}_i = 0$ once more,

$$\sum_{i=1}^{4}(XP_i)^4 = 4\Big(4 - 2\sum_{i=1}^{4}\mathbf{x}\cdot\mathbf{p}_i + \sum_{i=1}^{4}(\mathbf{x}\cdot\mathbf{p}_i)^2\Big) = 16 + 4\sum_{i=1}^{4}(\mathbf{x}\cdot\mathbf{p}_i)^2.$$

Ask yourself The linear terms vanished again. Why can't the same trick dispose of $\sum(\mathbf{x}\cdot\mathbf{p}_i)^2$?

A sum of squares is not linear in the $\mathbf{p}_i$, so knowing $\sum \mathbf{p}_i = \mathbf{0}$ says nothing directly about it. This is where the coordinates come in. With $\mathbf{x} = (x, y, z)$ and the vertices from part (ii),

$$\mathbf{x}\cdot\mathbf{p}_1 = z, \qquad \mathbf{x}\cdot\mathbf{p}_2 = \tfrac{1}{3}(2\sqrt{2}\,x - z),$$
$$\mathbf{x}\cdot\mathbf{p}_3 = \tfrac{1}{3}(-\sqrt{2}\,x + \sqrt{6}\,y - z), \qquad \mathbf{x}\cdot\mathbf{p}_4 = \tfrac{1}{3}(-\sqrt{2}\,x - \sqrt{6}\,y - z).$$

Square and add, collecting terms:

- x^2: $\tfrac{1}{9}(8+2+2) = \tfrac{4}{3}$; y^2: $\tfrac{1}{9}(6+6) = \tfrac{4}{3}$; z^2: $1 + \tfrac{1}{9}(1+1+1) = \tfrac{4}{3}$.
- xy and yz: these appear only in $\mathbf{x}\cdot\mathbf{p}_3$ and $\mathbf{x}\cdot\mathbf{p}_4$, which differ only in the sign of the y term, so they cancel.
- xz: $\tfrac{1}{9}\big(-4\sqrt{2} + 2\sqrt{2} + 2\sqrt{2}\big) = 0$.

Hence

$$\sum_{i=1}^{4}(\mathbf{x}\cdot\mathbf{p}_i)^2 = \tfrac{4}{3}\big(x^2 + y^2 + z^2\big) = \tfrac{4}{3},$$

because X lies on the unit sphere. Therefore

$$\sum_{i=1}^{4}(XP_i)^4 = 16 + 4\cdot\tfrac{4}{3} = \frac{64}{3},$$

which does not depend on the position of X, as required.

Ask yourself We used particular coordinates. Does the conclusion hold for every regular tetrahedron inscribed in the sphere?

Yes. Any such tetrahedron can be rotated about O so that P_1 goes to $(0, 0, 1)$ and P_2 into the half-plane $y = 0$, $x > 0$; rotations preserve lengths, and as X ranges over the sphere so does its rotated image. So the coordinates of part (ii) lose no generality.

Checks

- Take $X = P_1$. The distances are 0 and three edges, each of length $\sqrt{2 - 2(-\tfrac{1}{3})} = \sqrt{8/3}$. Then $\sum(XP_i)^2 = 3\cdot\tfrac{8}{3} = 8$ and $\sum(XP_i)^4 = 3\cdot\tfrac{64}{9} = \tfrac{64}{3}$, matching both results.

- Take $X = (0, 0, -1)$, the point opposite P_1. Then $XP_1 = 2$, and for the other three $\mathbf{x} \cdot \mathbf{p}_i = \frac{1}{3}$, so $(XP_i)^2 = \frac{4}{3}$. This gives $4 + 3 \cdot \frac{4}{3} = 8$ and $16 + 3 \cdot \frac{16}{9} = \frac{64}{3} \approx 21.33$.
- A computer check at thousands of random points on the sphere gave 8 and 21.333… every time.

Common slips

- Asserting that all the $\mathbf{p}_i \cdot \mathbf{p}_j$ are equal without saying why: it is the equal edge lengths, through $(P_iP_j)^2 = 2 - 2\,\mathbf{p}_i \cdot \mathbf{p}_j$.
- Forgetting that $|\mathbf{x}| = 1$, which is needed both for $(XP_i)^2 = 2 - 2\,\mathbf{x} \cdot \mathbf{p}_i$ and in the final step $x^2 + y^2 + z^2 = 1$.
- Arithmetic slips in the cross terms of part (iii): a sign error there leaves a spurious dependence on X.

Beyond STEP

The heart of part (iii) is the identity $\sum_i (\mathbf{x} \cdot \mathbf{p}_i)^2 = \frac{4}{3}|\mathbf{x}|^2$ for *every* vector $\mathbf{x}$. In matrix language, $\sum_i \mathbf{p}_i \mathbf{p}_i^{\mathsf{T}} = \frac{4}{3} I$: the four vertex vectors behave like an over-complete orthonormal basis, called a *tight frame*. This has to be true by symmetry, since the matrix commutes with every rotation of the tetrahedron and so must be a multiple of I, and its trace is 4. Point sets on a sphere whose low-degree averages match the sphere's own averages are called *spherical designs*; they are used in numerical integration and coding theory. For the linear algebra behind this (outer products, symmetric matrices and eigenvalues), see Strang, *Introduction to Linear Algebra*.

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